

Janitza UMG 96-EL SCADA Integration Guide

Extended Modbus Map, Data Acquisition Architecture, and Advanced Diagnostics

Device: Janitza UMG 96-EL Energy Analyser (Firmware v1.1.0+ / v1.2.0)

Document Reference: Extended Modbus Registry List & Mathematical Formulary

1. Data Acquisition & Protocol Execution Rules

Update Cycle Rate: The device internally recalculates and updates all core electrical parameter registries every **200 ms**.

Byte Order Sequence & Endianness: The network interface broadcasts raw data registers in **Big-Endian** architecture (high-byte preceding low-byte). For host configurations running native **Little-Endian** sequence mapping, an addressing offset factor of **32768** must be added to the baseline register coordinates during transmission queries.

Supported Function Codes: Standard Modbus compliance covers Slave Function Codes: **03** (Read Holding Registers), **04** (Read Input Registers), **06** (Preset Single Register), **16 / 10Hex** (Preset Multiple Registers), and **23 / 17Hex** (Read/Write 4X Registers).

Numerical Formats & Data Scaling Architecture:

- *float / int formats:* Automatically process native scaling logic based on calibrated physical transformer configuration ratios (CT / VT scales).
- *short format:* Bypasses device-side engineering conversion factor processing. Raw unscaled units are returned. Host application algorithms must run scalar multiplication algorithms matching the instrumentation ratios. Short integer tables are scaled internally by multiplier variables (e.g., x10, x100, x1000) to safely protect decimal significance; drivers must divide captured values by their prospective scalars.

2. Base Engineering Registers & Measured Values

Addresses	Type	Description	Unit	SCADA / Automation Allocation & Usage
19000	float	Voltage L1-N	V	Basic SCADA visualization & power quality tracking.
19002	float	Voltage L2-N	V	Basic SCADA visualization & power quality tracking.
19004	float	Voltage L3-N	V	Basic SCADA visualization & power quality tracking.
19012	float	Current I L1	A	Phase L1 loading telemetry & overcurrent monitoring.
19014	float	Current I L2	A	Phase L2 loading telemetry & overcurrent monitoring.
19016	float	Current I L3	A	Phase L3 loading telemetry & overcurrent monitoring.
19026	float	Sum Active Power (Psum3)	W	Total active power consumption for demand management; calculated as P1 + P2 + P3.
19034	float	Sum Apparent Power (Ssum3)	VA	Total apparent power baseline used for utility billing analysis; calculated as S1 + S2 + S3.
19042	float	Fund. Sum Reactive Power (Qsum3)	var	Fundamental reactive power for power factor correction (PFC) control logic loops; Q1 + Q2 + Q3.
19050	float	Measured Frequency (10s)	Hz	Grid stability analytics, frequency drift detection, and generation synchronization tracking.
19068	float	Real Energy L1..L3 Consumed	Wh	Primary accumulation register for SCADA billing profiles and historical energy dashboards.
19110	float	THD, U L1-N	%	Total Harmonic Distortion of voltage for asset health profiling and harmonic pollution baselines.
19116	float	THD, I L1	%	Total Harmonic Distortion of current to evaluate non-linear load stress and thermal heating risks.
10	float	Current Transformer Primary	A	Read/Write system configuration parameter to establish or verify primary scaling factors.
12	float	Current Transformer Secondary	A	Read/Write system parameter used to establish nominal secondary loop profile parameters.

3. Secondary Telemetry & Advanced Diagnostic Registers

The secondary memory structure of the Janitza UMG 96-EL contains hidden data structures critical for professional power quality indexing, vector analysis, and sequence calculation profiles.

Addresses	Format	Metric Parameter	Formulary & Technical Definition	SCADA Operational Diagnostic Application
5910	float	Crest Factor U L1	$C_f = U_{peak} / U_{rms}$	Measures transient voltage peak severity relative to standard wave envelopes to protect capacitor banking networks.
5912	float	Crest Factor U L2	$C_f = U_{peak} / U_{rms}$	Identifies acute crest clipping caused by heavy saturated switch-mode supplies on Phase 2.
5914	float	Crest Factor U L3	$C_f = U_{peak} / U_{rms}$	Tracks lightning strikes, commutation notches, or system spikes preceding hardware failure profiles.

802	float	Voltage Zero Sequence	$U_0 = 1/3 * (U1 + U2 + U3)$	Vectorial base metric indicating neutral displacement voltage stress profiles under load shifts.
804	float	Voltage Negative Sequence	Vector components with reversed phase rotation sequence.	Core parameter for true phase asymmetry metrics; identifies supply side imbalance.
806	float	Voltage Positive Sequence	Vector components with standard phase rotation sequence.	Tracks the absolute foundational integrity of the balanced working EM vector field.
920	float	Current Zero Sequence	$I_0 = 1/3 * (I1 + I2 + I3)$	Monitors earth return currents and path leakage profiles without isolated hardware sensors.
922	float	Current Negative Sequence	Reversed sequence vector summation matrix calculation.	Quantifies reverse magnetic rotating fields causing counter-torque heating inside motor stators.
924	float	Current Positive Sequence	Forward sequence vector summation matrix calculation.	Verifies productive balanced workload transformation components within connected motor networks.
---	Calculated	Voltage Imbalance Index	$Asymmetry \% = (U_{Geg} / U_{Mit}) * 100$	Triggers alarm interlocks for load asymmetry conditions, avoiding heavy insulation breakdown inside motor windings.
848, 850, 852	float	Voltage Vectors: Real Parts (L1, L2, L3)	$U_{real} = U_{rms} * \cos(\phi)$	Supplies raw Cartesian spatial mapping parameters used to compile dynamic power wheel diagrams.
854, 856, 858	float	Voltage Vectors: Imaginary Parts (L1, L2, L3)	$U_{imag} = U_{rms} * \sin(\phi)$	Provides phase displacement coordinate profiles mapped natively across complex engineering planes.
926, 928, 930	float	Current Vectors: Real Parts (L1, L2, L3)	$I_{real} = I_{rms} * \cos(\theta)$	Extracts real vector parameters mapping fundamental loading dynamics back into software loops.
932, 934, 936	float	Current Vectors: Imaginary Parts (L1, L2, L3)	$I_{imag} = I_{rms} * \sin(\theta)$	Isolates reactive flow component paths per phase, helping diagnose localized phase saturation issues.

4. Value Characterization & Integration Optimization

Register Duplication Layout: Energy accumulation metrics are mirrored across dual formatting tables. Active, reactive, and apparent energy variables exist as 32-bit Integer metrics alongside native 32-bit IEEE 754 Floating-Point structures. Floating-Point records are ideal for swift engineering calculations without external calibration shifts. Conversely, the standard 32-bit Integer registers avoid scaling noise and provide precise accumulator rollovers for PLC billing engines.

Spectral Log Analytics (Fourier Block Queries): The analyzer evaluates network frequencies up to the **40th harmonic block** for voltage and current waveforms. Industrial tracking routines should request sequential data blocks beginning at register address **1000** for standard 32-bit Floating-Point parameters, or address **3536** when deploying compact 16-bit Short Integer structures.

Native MQTT Architecture Mapping: To minimize communication overhead inside modern IoT cloud networks, the system supports an autonomous MQTT message architecture. Native mapping configurations route common data parameters directly to distinct topic payload tags without constant poll cycles. For instance, base telemetry register **19000** automatically maps into the ULNRms_L1 JSON payload element parameter string.

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